Chapter 3 Methodology Record: Maxwell and Icepak Simulations

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1. Model chain at a glance

This record covers how the electromagnetic and thermal numbers were obtained, not the numbers themselves (those are in the Results chapter). All Maxwell and Icepak work used **Ansys Electronics Desktop 2025 R2**, driven by Python (PyAEDT) scripts so that every model can be rebuilt from its configuration file. The two machines are **Rev0** (one 5-board stator between two rotor backplates) and **Rev1-S** (two 5-board stators and three rotor discs, stators in series).

No single model was both detailed enough and small enough to solve, so the work was organised as a chain in which each model was checked against the one before it:

#	Model	What it contains	What it produces	Role
M1	Detailed single-board FEM (magnetostatic)	Rotor (magnets, backplates, shaft) plus one PCB with every trace and via as copper	Torque per board (Lorentz and virtual work), per-board K	Reference for each board position
M2	Board-by-board sum	M1 repeated for boards 1–5	K of the stator as the sum of board K	Coreless superposition check
M3	Field map	Magnets and steel only; B sampled on 180 775 points along the real trace paths	K from the integral of the B-field along the copper (no copper meshed)	Fast check of M1/M2; used for the rotor-angle sweep
M4	Full-motor detailed static (HPC)	Whole Rev0 with all five boards in full copper detail	K of the complete machine	Final static reference

#	Model	What it contains	What it produces	Role
M5	FMROT rotating field-only (transient)	Magnets and steel rotating; stator zones as vacuum; no copper	Back-EMF (flux linkage on the trace paths), cogging, field animations, magnet eddy loss	Rotating results
M6	EQ equivalent coils (magnetostatic, then current sweep)	8 sector coils per phase per board, calibrated against M1	Torque against current to 3 A, armature-only inductance	Test C (current sweep)
M7	Icepak steady state	Boards, rotor, housing, surrounding air	Winding and magnet temperature against current	Thermal limit

Final (validated) runs used in the report: Rev0 M1–M5 complete (M4 = HPC job 1470376); Rev1-S M1–M3, M5 and M6 complete (the Rev1-S full detailed static, M4, was not obtained, see Section 7); M6 current sweeps HPC jobs 1504750 (Rev0) and 1504792 (Rev1-S); M7 Icepak v1 (Rev0) and v2 (Rev1-S). The torque-speed, efficiency and operating-point figures (Test D) are **Python post-processing** of these solved results, not further FEM solutions.

2. Maxwell 3D: solvers and materials

2.1 Solver types and settings

Setting	Magnetostatic (M1–M4, M6)	Transient (M5 FMROT)
Formulation	Magnetostatic, adaptive mesh	Transient, T-Ω formulation (A-Φ transient does not support motion in 2025 R2)
Adaptive passes	Max 4 at 20 % refinement (final profiles); config default 10; min 2	Not adaptive: fixed length-based mesh (Section 3.3)
Energy error target	1 % per pass	—
Linear residual	1e-6	—

Setting	Magnetostatic (M1–M4, M6)	Transient (M5 FMROT)
Nonlinear residual	0.001	0.005
Matrix solver	Direct (local runs); iterative on the HPC full motor (L2 ladder level)	Default
Time integration	—	Backward Euler (TimeIntegrationMethod 0)
Matrix (inductance)	Off; inductance from an armature-only energy solve instead	Off

Why these solvers. Torque at fixed rotor angles and the current sweep need no time dependence, so magnetostatic was used wherever possible: it is cheaper and its adaptive mesh gives a convergence measure. The transient solver was used only where rotation is essential (back-EMF waveform, cogging, eddy loss in the magnets). T-Ω was chosen because it is the transient formulation that supports a rotating band; it also allows eddy currents to be switched on per object.

Torque calculation. Two independent methods were computed in every M1 solve: (a) Lorentz force on the copper ($\int J \times B\, dV$ over the traces), and (b) virtual work on the rotor parts. Agreement between them was used as an accuracy check. K is defined by $T = 1.5 \cdot K \cdot \hat{I}$ (three-phase sinusoidal current, peak $\hat{I}$), so K equals the peak flux-linkage slope $k = d\lambda/d\theta$ used for the back-EMF.

2.2 Materials

Part	AEDT material	Properties used	Notes
Magnets (16 × 20×6×2 mm, axially magnetised, alternating)	NdFe35 (library)	Br ≈ 1.23 T, Hc ≈ −890 kA/m, μr ≈ 1.10, σ ≈ 0.625 MS/m	Linear permanent magnet; eddy currents on in M5
Backplates (Ø86/Ø19 × 2 mm) and shaft (Ø19)	steel_1010 (library)	Nonlinear B-H curve, σ ≈ 2 MS/m	Eddy currents off in M5 (DC field in the rotor frame)
Traces and vias	copper	σ = 5.8e7 S/m, μr = 1	Vias as 8-sided prisms, Ø0.30 mm

Part	AEDT material	Properties used	Notes
FR-4, solder mask, regions, zones	vacuum (air)	μr = 1, σ = 0	FR-4 treated as air: it is non-magnetic and non-conducting; a check with FR-4 included changed K by ⩽ 0.22 %

The magnet and steel values are the AEDT 2025 R2 library defaults; they were not replaced by supplier data for the purchased N35 magnets or the actual backplate steel (an assumption listed in Section 7).

3. Maxwell 3D: geometry, boundaries and mesh

3.1 Geometry included or excluded

Common to all models: 16 magnets on r = 28 mm, backplates Ø86/Ø19 × 2 mm, 1.0 mm mechanical air gap each side, 1.0 mm boards (5.10 mm 5-board stack with mask), shaft on the rotor side only (5 mm above the rotor, united with the backplate). Rev1-S: two stators, three rotor discs; the rear steel was set 1.0 mm behind its magnets (REV1_S2_ENV).

Model	Included	Excluded (and why)
M1 detailed board	Rotor; one board's traces and vias (scaled stack-up, 35 μm copper, mask 10.16 μm treated as air)	Other four boards (coreless: superposition holds, Section 5.2); housing, bearings, fasteners (non-magnetic)
M3 field map	Rotor; vacuum annulus r 13.5–47.5 mm in each stator layer for sampling	All copper (B is the magnet field; winding field below 3 mT, Section 5.2)
M4 full motor	Rotor and all five boards in full detail	Housing, bearings, fasteners
M5 FMROT	Rotor (moving); stator zones as vacuum	Copper (back-EMF from flux on the trace paths, so no conductor mesh is needed)
M6 EQ coils	Rotor; 8 thin sector coils per phase per board (slab 0.08 mm, arc width 1.5 mm, side trim 0.8 mm)	Detailed traces, replaced by calibrated coils

3.2 Boundary conditions and air region

The outer region boundary is the AEDT default zero tangential H (natural) condition, i.e. flux leaves normally. No symmetry or periodic boundaries were used: the three-phase winding and lead-outs are not periodic per pole pair, and a full model avoided a source of error.

The region is a box set by a **padding** around the model. Final runs used **20 mm** padding on every side except the lead-end (+Y) face of the full motor, padded 0 because the leads leave the board there. Padding 20 mm is about 1/4 of the rotor diameter and 10 times the air gap. Justification: increasing padding from 5 to 20 mm raised board K by 3.1 % (board 1) and 3.9 % (board 5), while the change beyond 20 mm was within the convergence error; the early 5 mm results (K 18.05/18.02 mN·m/A) were therefore superseded. For the Rev1-D comparison model, 20 mm made the model too large and 5 mm was used (stated wherever quoted).

3.3 Mesh settings and convergence

Item	Setting
Air gap (magnetostatic)	Max element length 0.5 mm (= gap/2), capped at 300 k elements
Magnets	Max length 2.0 mm (static); 1.0 mm (M5)
Field-map / FMROT stator zones	0.8 mm, capped at 500 k elements
Rotating band (M5)	1.0 mm; band clearance 0.5 mm from moving parts
Steel (M5)	2.5 mm
EQ coils	1.5 mm
Adaptive refinement	4 passes × 20 %, 1 % energy-error target (full motor on HPC: 6 passes, iterative solver)

Mesh independence was checked, not assumed: refining the field-map zone from 0.8 to 0.55 mm changed K by 0.03 %; a gap mesh of 1.0 against 2.5 mm changed it by 0.01 %; doubling and reversing current (linearity) changed K per amp by 0.002 %. The transient model is not adaptive, so its mesh was fixed from these magnetostatic checks, and its torque and EMF were then compared with the static results (agreement within 1.5 %).

Element counts recorded: detailed board model 1.93 M tetrahedra (10.4 GB RAM). The full-motor element count was not written to the HPC summary (field left blank) and is reported only as memory: 109.7 GB peak.

4. Maxwell 3D: excitation, winding and motion

4.1 Excitation and winding connection

- **Detailed models (M1, M4):** current excitation on the phase terminals at the board lead-outs; traces and vias are solid copper, so the current distributes itself (current density is a result, not an input). Each board's phase is excited separately so per-board and per-phase torque can be read.
- **Per-board characterisation:** single phase (phase A) at +I and −I (`SWEEP_MINUS_I`): the difference removes the magnet-only torque and gives the current-proportional torque cleanly. Board results are combined as phasors, $K = |\Sigma K_k \cdot \exp(j\psi_k)|$.
- **Three-phase synchronous check:** $i_X = \hat{I} \cdot \cos(p\theta + \varphi_{sync} + \varphi_X)$, $p = 8$, $\varphi_A = 0°$, $\varphi_B = -120°$, $\varphi_C = +120°$, $\hat{I} = 1$ A. The phase windings are the same pattern rotated by 0, −30 and −60° mechanical; the script refuses to run if the electrical offsets do not match these rotations ($p \cdot \Delta\theta_{mech} = \Delta\varphi \bmod 360°$).
- **Board and stator connection:** the five boards of a stator are in series per phase. Rev1-S: two stators in series in the final models; parallel connection was evaluated in post-processing only. Star connection is assumed for all line quantities (the bench tests are in star).
- **EQ coils (M6):** each phase is 8 stranded sector coils per board, driven with $N_{coil} \cdot I$. N_{coil} was calibrated per board against the detailed K (EQ_DETAILED_K), because geometric turns over-predict K by 24 % (Rev0 turns factor 0.809, board spread 7.1 %). Current sweep: $\hat{I}$ = 0, 0.5, 1, 1.5, 2, 3 A at the calibrated current angle, $i_X = N_{coil} \cdot \hat{I} \cdot \cos(p\theta - \psi_X)$, each at rotor angles 0° and 3.75° (30° electrical apart, so the mean cancels the sixth-harmonic ripple); then one armature-only solve (magnets off) for inductance, $L - M = W/(0.75 \cdot \hat{I}^2)$.

4.2 Rotor-angle conventions

16 poles, so 45° mechanical = one electrical period and 1° mech = 8° elec. $\theta = 0$ is the rotor position built in the CAD (magnet 1 centred on +X); positive θ is counter-clockwise about +Z. Magnetostatic sweeps used: full period 0–45° in 5.625° steps (45° elec, 9 angles) for the reference sweep; 0–16.875° (4 angles) for design-of-experiments runs; field map 0–39.375° (8 angles, one period). Half-wave antisymmetry ($T(\theta + 22.5°) = -T(\theta)$ at reversed current) was checked and used to shorten some sweeps and to fill the corrupted Rev1-S calibration set (Section 7).

4.3 Motion (transient M5)

Item	Setting
Motion	Rotating band about Z, non-cylindrical off; one band (Rev0), two bands (Rev1-S, with the middle rotor steel split)
Speed	1000 rpm constant (f_e = 133.3 Hz); other speeds from linear scaling, checked by Test D
Time step	1/24 of an electrical period = 0.3125 ms (FMR_SPP = 24; 48 in the base config)
Duration	One electrical period (7.5 ms, 24 steps); 6-step SHORT runs used first for timing and validation
Initial condition	Magnets only (no current); load torque from the field on the trace paths at Î = 1 A in post-processing
Eddy currents	On in magnets, off in steel

The time step was set by disk, not by accuracy: field files at every step cost about 0.5 GB each, so 48 steps per period would exceed the 45 GB quota guard. With 24 steps the 8th-harmonic content of the EMF is resolved (Nyquist at the 12th), and the EMF harmonic results (THD) were checked against the 8-angle field map.

5. Solutions completed, validation and computing resources

5.1 Solutions completed per configuration

Solution	Rev0	Rev1-S	Where solved
M1 detailed board, boards 1–5, ±I, rotor-angle sweep	Complete	Complete (per stator)	Lab PC
M3 field map, 8 angles	Complete (FIELDMAP_REV0_P20)	Complete (FIELDMAP_REV1S_P20)	Lab PC

Solution	Rev0	Rev1-S	Where solved
M4 full-motor detailed static, three-phase synchronous, two rotor angles	Complete (job 1470376)	**Not obtained** (job 1486831 stopped by the scratch quota after 48 h, before one angle finished)	UCT HPC
M5 FMROT transient, 1 period at 1000 rpm	Complete	Complete (two bands)	UCT HPC
M6 EQ calibration	Complete	Complete (offline, antisymmetry fill)	UCT HPC
M6 EQ current sweep 0–3 A + armature-only	Complete (job 1504750)	Complete (job 1504792)	UCT HPC
M7 Icepak steady state, rotor still	Complete (0.737, 1, 2 A fans on; 2 A fans off)	Complete	Lab PC

5.2 Validation methods

Check	How	Outcome
Single-board gates (4–6 Oct)	DC resistance from the ohmic-loss integral; inductance by flux linkage and by induced voltage; three phases together; FR-4/mask as air; ±I antisymmetry; 0–45° periodicity; phases B, C against A	R 0.2778/0.2774/0.2779 Ω (0.19 % spread); L ≈ 2.07 µH both routes (within 1 %); FR-4 ⩽ 0.22 %; antisymmetry 0.002 %; periodicity 0.02 %; K of B, C within 0.31 % of A
Lorentz against virtual work	Copper torque and rotor torque from the same solve	Within 0.05–0.5 % (full motor 18.534 against 18.470 mN·m/A)
Board-by-board sum against full motor (Rev0)	Σ board phasors (M2) against M4	18.540 against 18.534 mN·m/A (+0.03 %)

Check	How	Outcome
Field-only against detailed conductor	M3 against M1 per board	+1.4 % at board 1 rising to +10 % at board 5; +5 % for the stator. Attributed to the filament (centre-line) approximation; mesh, boundary and shaft geometry ruled out (each ⩽ 0.1 %). The field map is therefore used only as a consistent upper bound and for shapes and angles, never for a headline K
Rotating against static	M5 against M3 at every angle and phase	Within 1.49 % (Rev0), 1.59 % (Rev1-S)
Equivalent coils against detailed	Calibration (turns factor 0.809 for Rev0; geometric turns over-predict K by 24 %); Rev0 calibration applied unchanged to Rev1-S	Rev1-S 39.30 against 39.50 mN·m/A (−0.5 %); sweep at 1 A gives 18.52 mN·m/A (−0.08 % against M4)
Linearity	Steel at 2 A; current sweep 0.5–3 A	k changed 0.002 %; K changed +0.008 % (Rev0), +0.004 % (Rev1-S)
Mesh independence	Section 3.3	⩽ 0.1 %

The coreless assumption behind the board-by-board sum and the field-only model was itself tested: the winding field at 3 A (1.3–2.6 mT) is under 0.6 % of the magnet field and the steel flux density did not change between 0 and 3 A, so superposition holds and the trace copper does not need to be in the rotating model.

5.3 HPC resources

- **Cluster:** UCT HPC, SLURM, partition *ada*, account *eleceng*; AEDT 2025 R2 in batch (non-graphical) mode via `hpc_motor.py`.
- **Runner logic:** a supervisor checks the disk (stops jobs at a 45 GB guard on the shared 50 GB scratch quota), writes an .acf file with cores and 85 % of the allocated RAM, and steps through a memory ladder L0 (direct, 8 passes) → L1 (iterative) → L2 (iterative, 6 passes, 20 %, 1 % error) → L3–L5 (coarser) after an out-of-memory failure. Point-by-point runs resume after the last finished point. All scripts were first exercised against a mock AEDT harness.

- **Rev0 full motor (job 1470376):** 32 cores, 243.8 GB allocated, ladder level L2; 1036 and 1047 min for the two rotor angles (about 17.5 h each), peak 109.7 / 109.3 GB; average CPU use about 2.9 of 32 cores, i.e. memory-bound.
- **Rev1-S full motor (job 1486831, superseded):** 40 cores, 250 GB peak, 48 h, no angle completed.
- **Rotating models (M5):** Rev1-S needed 32 cores for its 122 GB peak (about 7.6 GB per core available).
- **Test C sweeps:** job 1504750 (Rev0, 16 cores) and job 1504792 (Rev1-S, 32 cores), 24 h limit; each point's solution deleted after read-out, so the disk stayed small (266 MB when checked).
- **Lab PCs (16 GB):** single-board models (1.93 M tetrahedra, 10.4 GB), field map (0.8 min per angle), Icepak.

6. Icepak thermal model

Icepak (AEDT 2025 R2) steady-state conjugate heat transfer, built by `REV0_ICEPAK_BUILD.py` (v1, 8 Oct) and `REV1_ICEPAK_BUILD.py` (v2), and solved by `run_REV0_ICEPAK_STEADY.py` / `run_REV1_ICEPAK_STEADY.py`. Build and solve are separate scripts: the build saves a validated "frozen" project and each solve works on a copy.

6.1 Geometry

- **Stator:** each board is a 12-gon (apothem 46.5 mm, bore Ø25 mm, 1.0203 mm thick). Rev0: each board split into a winding annulus *Win_k* (r 16–40 mm, where the copper loss is applied) and the rest of the board *Board_k*; boards touch with no contact resistance. Rev1-S (v2): each stator modelled as one merged 5-board body plus one winding annulus, with per-board point monitors. Layout S1 at z = 20.000 mm, S2 at 31.102 mm, top of R2 steel 41.203 mm.
- **Rotor:** steel disc Ø86/Ø19 × 2 mm, 16 magnets 20×6×2 mm, 1.0 mm gap; **stationary** (no rotation, no rotor stirring).
- **Shaft and bearings:** Ø19 steel shaft through both end plates; each bearing an effective ring sized for about 0.3 W/K.
- **Housing (assumed, not measured):** sealed acrylic tube ID 120 mm × 3 mm wall, 60 mm internal length; 140 × 140 × 6 mm aluminium end plates; aluminium base plate 160 mm, 6 mm; 6 × M3 steel tie rods on a 90 mm PCD; stack bottom 20 mm from the plate.

6.2 Materials

Material	k (W/m·K)	ρ (kg/m³)	cp (J/kg·K)	Used for
PCB_eff (anisotropic)	in-plane 40.3, through-plane 0.334	2600	900	Boards and windings: 6 × 35 µm copper at 50 % coverage in 1.02 mm FR-4 (copper fraction 10.3 %); in-plane by parallel mixing (Cu 385, FR-4 0.8), through-plane by series mixing (FR-4 0.30)
Steel1010_th	52	7870	460	Rotor discs, shaft, rods
NdFeB_th	9	7500	420	Magnets
Acrylic_th	0.19	1190	1470	Tube
Al6061_th	167	2700	900	End and base plates
Bearing_eff	3.6	7800	460	Bearing rings

Densities and heat capacities do not affect a steady-state result; they are listed for completeness.

6.3 Boundary conditions and heat transfer

- **Enclosure (computational region):** padding +X/−X/+Y 100 mm, ±Z 150 mm, −Y 0 (the bench). Axes: Z = shaft = fan direction, Y = up.
- **Inlet:** −Z face, free opening with velocity 1 m/s at 30 °C (bench fume hood with fans on). Other open faces: pressure openings at 30 °C. −Y bench: adiabatic wall.
- **Physics:** flow and heat solved together; laminar; gravity −Y (natural plus forced convection); radiation off (conservative, gives higher temperatures).
- **Fans-off case:** inlet velocity 0, natural convection only.

6.4 Heat source and R(T)

Copper loss only (eddy loss is under 1 % of copper loss at 1000 rpm and was added later in Test D post-processing). Per board: $P = 1.5 \cdot \hat{I}^2 \cdot R20 \cdot (1 + \alpha(T - 20))$, R20 = 0.2777 Ω per board-phase (from the Maxwell DC gate), α = 0.00393 /K, T = the mean winding temperature from the previous solve. The loop starts at 20 °C and repeats until the loss changes by less than 1 % (4–5 iterations at 2 A). Cases: Rev0 0.737, 1 and 2 A pk fans on, 2 A fans off; Rev1-S 0.345 A (20 mN·m), 1 and 2 A fans on, 2 A fans off.

6.5 Solver, mesh and convergence

- Steady state, maximum 400 iterations per solve, default Icepak residual criteria; about 2.5–3 min per solve on a lab PC.
- Mesh: global maximum element 8 mm, minimum gap 0.25 mm, 2 buffer layers; local mesh region on the stator stack 2 × 2 × 0.5 mm (finer through the thickness, where the temperature gradient is).
- Rev1-S: the HD mesher had dropped one winding when each of 20 thin bodies was separate; merging each stator into one body, setting mesh priorities (windings above boards) and a scripted mesh check ("objects without mesh: none") fixed this. AEDT Validation Check passed for every frozen build.
- No separate mesh-refinement study was run for Icepak (a limitation, Section 7).

6.6 How the model was checked

- **Energy:** the assigned loss was read back for each case; each converged loss matches $1.5 \cdot \hat{I}^2 \cdot R(T)$ at the reported winding temperature.
- **Independent read-back:** `check_REV0_ICEPAK_TEMPS.py` re-reads mean and maximum temperature of every body from the solved project (this found and fixed two read-out bugs: magnet volume maximum, and a Kelvin/Celsius conversion).
- **Lumped network:** `rev0_thermal_lumped.py`, an independent hand-built thermal network, was calibrated to Icepak (bearing, hub and cavity coefficients) and then reproduced the other Icepak cases within about ±7 %.
- **Physical sense:** board temperatures rise toward the middle of the stack; fans-off is hotter than fans-on; Rev1-S per stator is hotter than Rev0 because its stators share cooling surfaces.
- Rev1-S per-board point monitors returned no value in the fans-on cases; the winding-body mean and maximum were used for all Rev1-S results.

6.7 Differences from the manufactured motor

- Housing dimensions and materials are assumed (no drawing was available when built).
- Rotor still: no rotor-driven air circulation (pessimistic for running conditions).
- Board-to-board contact perfect (a contact coefficient of 100–2000 W/m²K changes board temperature by under 3 K in the lumped model).
- Copper as a smeared effective material, not individual traces; hot spots inside a board are not resolved.
- Radiation off; ambient 30 °C and 1 m/s hood flow assumed.
- Resistance from the Maxwell model (0.2777 Ω per board-phase), not the measured value.

7. Reproducibility and limitations

7.1 Status of each value

Category	Values
Completed FEM/CFD solutions	Per-board K and phase (M1); Rev0 full-motor K (M4); field maps (M3); back-EMF waveforms, harmonics, cogging, steel flux density and magnet eddy loss at 1000 rpm (M5); torque against current 0–3 A and armature-only energy (M6); Icepak temperatures at the listed currents (M7); DC resistance and single-board inductance gates
Post-processing of solved results (Python)	Rev1-S K as the board-by-board sum; inductance L − M from the energy; eddy loss at other speeds (scaled with f^2, checked by skin depth); torque-speed envelopes, efficiency maps and operating points (Test D); thermal resistance K/W and the 60 °C currents; Rev1-S parallel-connection estimates
Estimates	The cautious thermal basis (15/8.75 scaling); mechanical loss; the Rev1-S full-motor accuracy (±1 %, inferred from two methods agreeing within 0.5 %)
Assumptions	Library magnet and steel properties; housing geometry; 30 °C ambient and 1 m/s flow; 1.0 mm air gaps and 5.10 mm stack as designed (not as built); star connection; sinusoidal current in phase with the EMF

7.2 Scripts and configuration files

File	Purpose
`lab_v10/config.py`	Every geometric, material, mesh and solver setting, and the named run profiles (REF_ROTOR_MS_SWEEP, FIELDMAP_*, *FMROT_REV0/REV1S, EQ_CAL, EQ**_ISWEEP)

File	Purpose
`lab_v10/build_rotor.py`, `build_pcb.py`	Rotor and PCB copper geometry (from the Gerber-derived trace data)
`lab_v10/setup_model.py`	Region, boundaries, excitations, solver setups, sweeps and reports
`lab_v10/fm_rotating.py`, `transient.py`	M5 rotating model: zones, bands, motion, fixed mesh
`lab_v10/eqcoil.py`	M6 equivalent coils, calibration and current sweep
`lab_v10/check_traces_analytic.py`	Analytical check of the trace geometry
`build/src/supervisor_template.py`, `hpc_motor.py`	HPC runner: SLURM header, disk guard, memory ladder, resume
`HPC_RUN_STEPS_TestC.txt`	Exact commands for the Test C jobs
`Thermal/REV0_ICEPAK_BUILD.py`, `REV1_ICEPAK_BUILD.py`, `fix_REV1_ICEPAK_MESH.py`, `fix_REV1_ICEPAK_FROZEN.py`	Icepak builds and mesh fixes
`Thermal/run_REV0_ICEPAK_STEADY.py`, `run_REV1_ICEPAK_STEADY.py`, `check_REV0_ICEPAK_TEMPS.py`, `rev0_thermal_lumped.py`	Icepak solves with R(T), read-back check, lumped network
`testsC/current_sweep.py`, `testsD/speed_sweep.py`, `spec/ratings4.py`	Post-processing for Tests C, D and the operating points

These are collected in the evidence package sent with this record.

7.3 Numerical settings that materially changed results

Setting	Change	Effect	Final choice
Region padding	5 → 20 mm	K +3.1 % (board 1), +3.9 % (board 5)	20 mm (5 mm only for the Rev1-D comparison)
Field-map method	Filament (centre-line) against volume force	+1.4 % to +10 % per board, +5 % stator	Field map used only for shapes and as an upper bound

Setting	Change	Effect	Final choice
EQ coil geometry	Overlapping coil slabs (0.26°)	Mesh failure	Sides trimmed 0.8°, slabs 0.08 mm
EQ turns	Geometric against calibrated	+24 % K	Calibrated (0.809)
Point-file units	mm read as metres	All rotating field values NaN	Export probe accepts only finite results; re-run post-processing
Transient time step	48 → 24 steps per period	Disk use halved; harmonics up to the 11th still resolved	24
Icepak meshing (Rev1-S)	20 separate thin bodies	One winding not meshed	Merged stator bodies, mesh priorities, mesh check

Settings that did **not** change results (checked): mesh refinement (⩽ 0.1 %), FR-4 as air (⩽ 0.22 %), shaft stub geometry (⩽ 0.02 %), backplate 2 → 5 mm (+0.1 %), current level (0.002–0.008 %).

7.4 Final runs against superseded or failed attempts

Attempt	Status	Replaced by
Three full-motor solves on the 16 GB lab PCs	Failed in meshing / first pass	Board-by-board method; HPC job 1470376
Detailed T-Ω transient on the lab PCs	Impractical (25 min per phase-step, two freezes)	Magnetostatic angle sweeps and the field-only rotating model
K = 18.05 / 18.02 mN·m/A (5 mm padding)	Superseded	20 mm padding; final K 18.53
Rev1-S full detailed static (job 1486831)	Lost after 48 h (quota)	Board-by-board sum + EQ model (agree within 0.5 %)
EQ_REV1S_CAL project	Corrupted	Calibration completed offline using half-wave antisymmetry
Rotating runs at 48 steps per period	Killed at the 45 GB guard	24 steps per period

Attempt	Status	Replaced by
Neumann (analytical) inductance 110 µH	About 4× too high	Maxwell armature-only energy, 34–51 µH per stator
1.75 A pk thermal limit	Withdrawn	Icepak-based 60 °C currents (Test D)
Rev1-S Icepak v1 (10 separate boards)	Mesh dropped a winding	v2 merged model

7.5 Main limitations

- No full-motor detailed solve for Rev1-S.
- Library material data, not supplier data, for N35 and the backplate steel.
- The transient model has no conductors, so AC copper effects come from analysis (skin depth), not from FEM; circulating currents between parallel tracks are not modelled.
- Icepak: assumed housing, still rotor, no mesh-refinement study, radiation off.
- All results are for the designed geometry; the as-built gaps and stack are treated in the Discussion.